

AI FOR VALIDATION

CHAPTER 9

From Discovery to Defensible Findings

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Why Discovery Results Need a Statistical Test

Search keeps the strongest-looking candidates. That makes the discovery estimate optimistic.

BOOK SIMULATION

32 candidates

All true effects = 0

20,000 repetitions

REUSE DISCOVERY DATA

Selected candidate:
+2.073 SE

False confirmation:
56.1%

USE RESERVED DATA

Same selected candidate:
0.000 SE on average

False confirmation:
2.3%

Declared rate: 2.5%

Discovery finds the candidate. Confirmation tests whether the effect survives new evidence.

Choose the Test From What You Measured

The outcome and dependence structure determine the method.

Binary failure rate	Wrong / acceptable answer	Exact binomial + Clopper–Pearson
Continuous loss	Brier loss, loss inflation	Bootstrap interval
Repeated task variants	Same fact under several levels	Fact-clustered bootstrap
Factor effects	Anchor, pressure, alias	Logistic regression / deviance
Several findings	Multiple finalists	BH/FDR or family-wise control
Choose the method from the measurement, the sampling unit and the dependence in the data.		

Method 1 — Is the Failure Rate Above a Material Threshold?

Example: can we conclude that the underlying failure rate exceeds 20%?

1. STATE THE HYPOTHESIS

Let p be the underlying failure rate and τ the material threshold.
 $H_0: p \leq \tau$ $H_1: p > \tau$

Use a one-sided test because the claim is that failure is too high.

2. COMPUTE A LOWER BOUND

For k failures among n independent held-out cases:
 $p_L = \text{Beta}^{-1}(\alpha; k, n - k + 1)$

Confirm $\iff p_L > \tau$

3. SEE WHAT SAMPLE SIZE DOES

Observed failure rate is 100% in every row:

2 / 2 $\rightarrow$ 0.224

5 / 5 $\rightarrow$ 0.549

10 / 10 $\rightarrow$ 0.741

Same rate. Much stronger evidence as n grows.

The lower bound must clear the material threshold — not merely the observed rate.

Method 2 — Continuous Loss: Bootstrap the Effect

For Brier loss or loss inflation, keep the size of the error instead of turning every case into pass/fail.

1. DEFINE THE EFFECT

Example from the book:

$$s = \frac{\bar{L}_{\text{region}}}{\bar{L}_{\text{population}}} - 1$$

$s > 0$ means the selected region has higher mean loss than the population.

2. RESAMPLE HELD-OUT ROWS

Sample held-out rows with replacement.

Keep each row's region membership and loss together.

Recompute s for every bootstrap sample.

3. BUILD THE INTERVAL

Repeat, for example, 2,000 times.

Take bootstrap quantiles as the confidence interval.

For a material threshold δ :

confirm when lower bound $> \delta$

Resample the independent held-out unit and recompute the statistic each time.

Method 3 — Repeated Prompts: Resample the Underlying Facts

Five rewrites of one fact are five responses, but not five independent facts.

THE DATA

Fact A
clean • anchor •
pressure

Fact B
clean • anchor •
pressure

Fact C
clean • anchor •
pressure

Keep each fact's variants

FACT-CLUSTERED WEAK-NULL BOOTSTRAP

1. Compute each fact's mean failure at each factor level.
2. Center the levels to a common mean under H_0 .
3. Resample facts with replacement.
4. Recompute the spread across factor levels.

TURN IT INTO A p-VALUE

Compare the observed spread with the null-bootstrap spreads.

p-value = fraction of null resamples with a spread at least as large as observed.

Small p → evidence that factor levels differ.

Cluster at the independent unit. Repeated prompts do not create new independent tasks.

Method 4 — Which Factors Are Associated With Failure?

Use the designed factors from Chapter 8 to estimate main effects and interactions.

BINARY OUTCOME

For each executed test:

$Y = 1$ failure

$Y = 0$ success

Fit a logistic model to the per-test outcomes.

MAIN EFFECTS

Example model:

$$\text{logit } P(Y = 1) = \beta_0 + \beta_1 A + \beta_2 P + \beta_3 E$$

Analysis of deviance asks whether adding a factor improves the fitted model.

INTERACTIONS

Add an interaction term:

$$+ \beta_{12}(A \times P)$$

This asks whether the effect of anchoring changes when confidence pressure is also present.

Main effects ask which factors matter. Interactions ask whether combinations matter differently.

Method 5 — Multiple Findings: Control the Family Error

If several finalists are tested, a raw p-value below 0.05 may not survive the portfolio decision.

Rank	Raw p	BH cutoff	Adjusted p	Decision
1	0.004	0.010	0.0200	Reject
2	0.018	0.020	0.0350	Reject
3	0.021	0.030	0.0350	Reject
4	0.049	0.040	0.0613	Retain
5	0.200	0.050	0.2000	Retain

BENJAMINI-HOCHBERG

Order the p-values. At rank j, compare:
$$p_{(j)} \leq \frac{j}{M} \alpha$$

Keep the largest rank that qualifies and reject through that rank.

Here raw p = 0.049 does not survive.

BH controls false discovery rate. Use family-wise control when that is the stated objective.

Chapter 9 Statistical Toolkit

Match the statistical method to the evidence you actually have.

Failure rate > threshold

Exact binomial / Clopper–Pearson

Continuous loss

Bootstrap interval

Repeated task variants

Fact- or task-clustered bootstrap

Factor effects

Logistic regression / analysis of deviance

Interactions

Logistic model with interaction terms

Many hypotheses

BH/FDR or justified family-wise control

The method follows the measurement, the sampling unit and the dependence in the data.